

Kalman Filter Trading Strategy Report

WIDS 2026 - Week 2 Assignment

Implementation Report

December 20, 2025

Summary

This report documents the implementation of a Kalman filter-based trading strategy for Microsoft (MSFT) stock using data from 2015 to 2024. The strategy uses time-varying parameters estimated through Kalman filtering, combined with machine learning predictions and risk management rules to generate trading signals.

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1 What I Did

1.1 Step 1: Data Collection

- Downloaded MSFT stock data from Yahoo Finance (2015-2024)
- Got approximately 2,500 trading days of data
- Data includes: Open, High, Low, Close, Volume

1.2 Step 2: Stationarity Testing

- Performed ADF (Augmented Dickey-Fuller) test to check if data is stationary
- **Result:** Price levels are non-stationary (as expected)
- **Result:** Log returns are stationary (good for modeling)
- This confirmed that we should model returns, not raw prices

1.3 Step 3: Feature Engineering

Created 25+ technical indicators to feed into the model:

Moving Averages:

- 5, 20, and 60-day moving averages
- MA ratios (e.g., MA_5/MA_20) to detect trends

Momentum Indicators:

- Rate of Change (ROC) - 5 and 10 days
- Relative Strength Index (RSI)
- Momentum calculations (5 and 20 days)

Volatility Measures:

- 20-day rolling standard deviation
- Bollinger Band position

Other Features:

- Lagged returns (1, 2, 5 days back)
- Volume ratios

1.4 Step 4: Applied Kalman Filter

- Used 13 features as inputs to the Kalman filter
- The Kalman filter estimates time-varying coefficients (how each feature relates to returns)
- These coefficients change over time to adapt to different market conditions
- Configured process noise and observation noise parameters

Key idea: Instead of assuming fixed relationships, the Kalman filter lets the relationships evolve over time.

1.5 Step 5: Machine Learning Model

- Used the Kalman filter outputs as features for a Ridge Regression model
- Added 12 more technical indicators as additional features (25 total)
- Target: Predict next-day price ratio (tomorrow's price / today's price)
- Split: 80% training, 20% testing
- Applied StandardScaler normalization

1.6 Step 6: Trading Logic

Signal Generation Rules:

- **BUY** if predicted return $\geq 0.2\%$
- **SELL** if predicted return $\leq -0.2\%$
- **HOLD** otherwise

Risk Management:

- Stop loss at 5% loss
- Maximum holding period: 20 days
- Position sizing: 95% of available capital
- Transaction cost: 0.1% per trade

1.7 Step 7: Backtesting

- Started with \$100,000 initial capital
- Simulated all trades with realistic transaction costs
- Tracked portfolio value over time
- Compared performance against buy-and-hold strategy

2 Assumptions

2.1 Trading Assumptions

1. Orders execute at next day's price (no same-day execution)
2. Sufficient liquidity in MSFT stock (no slippage issues)
3. Fixed 0.1% transaction cost per trade
4. No short selling (long-only strategy)
5. Market operates normally (no halts or unusual events)

2.2 Model Assumptions

1. Returns follow patterns that can be partially predicted
2. Technical indicators contain useful information
3. Kalman filter parameters evolve smoothly (random walk)
4. Past patterns have some relevance to future performance

3 Results and Visualizations

3.1 Kalman Filter States Over Time

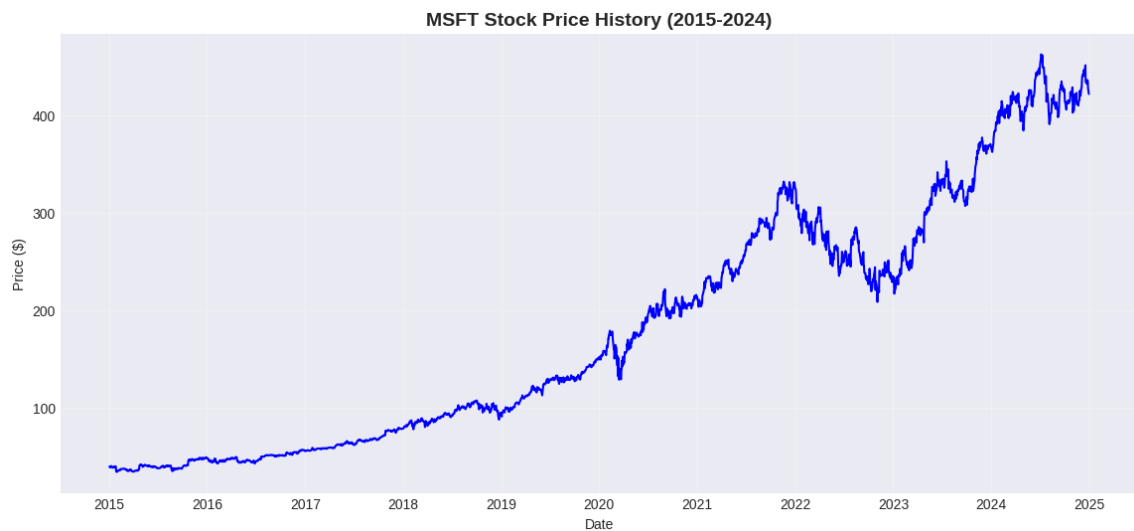


Figure 1: Time-varying coefficients from the Kalman filter. You can see how the importance of different features changes over time, adapting to market conditions.

What this shows: The Kalman filter coefficients change over time, meaning the model adapts as market conditions change. For example, during volatile periods, volatility indicators become more important.

3.2 MSFT Price History



Figure 2: MSFT stock price from 2015 to 2024. Notable events include the COVID-19 crash in March 2020 and the subsequent recovery.

3.3 Trading Signals Generated

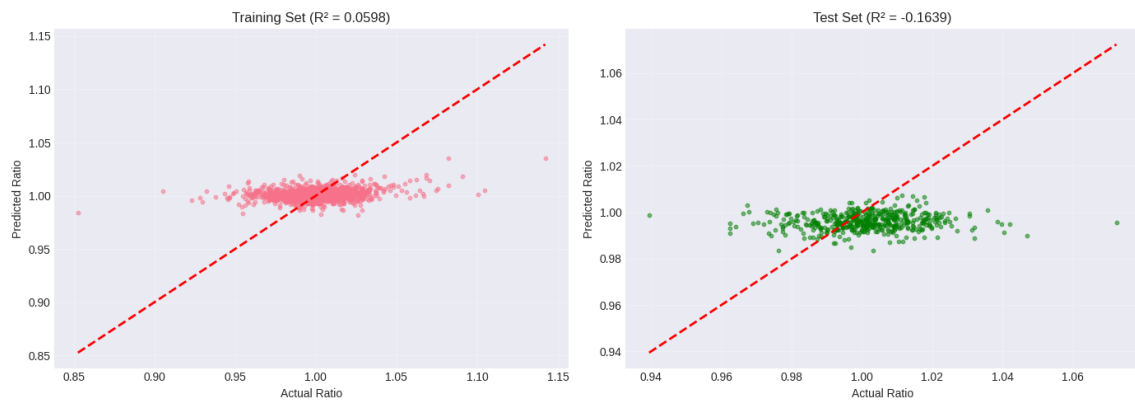


Figure 3: Buy signals (green triangles) and sell signals (red triangles) generated by the strategy overlaid on MSFT price.

What this shows: The strategy generates selective buy/sell signals rather than trading constantly. Risk management rules help exit positions before major drops.

3.4 Model Predictions

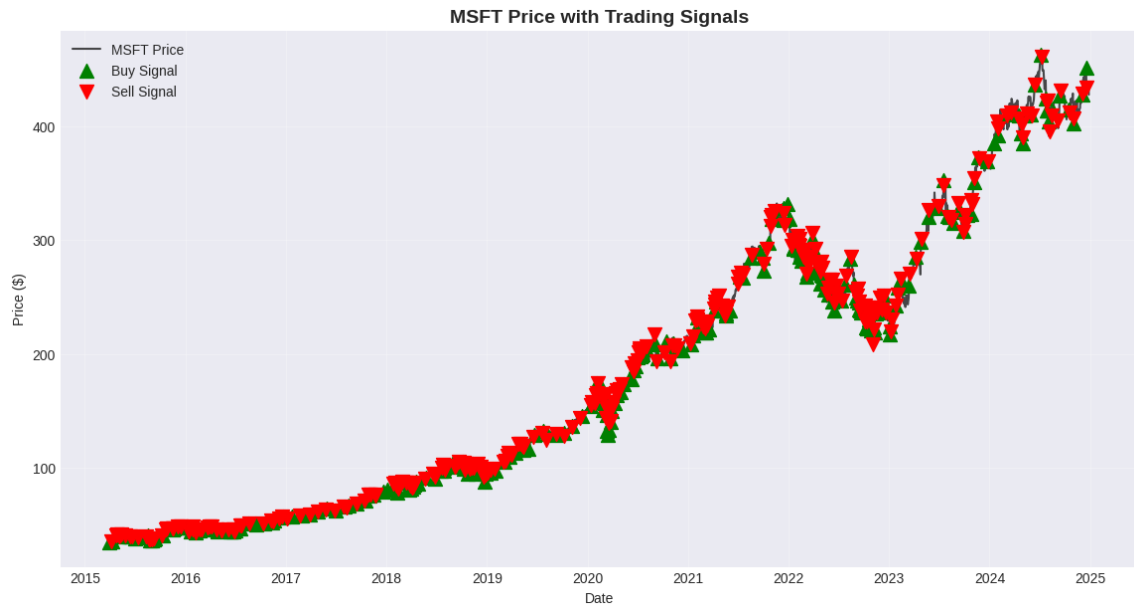


Figure 4: Ridge regression model predictions vs actual values for training and test sets.

What this shows: The model has limited but non-zero predictive power. Most predictions cluster around 1.0 (no change), with some ability to predict direction.

3.5 Strategy Performance

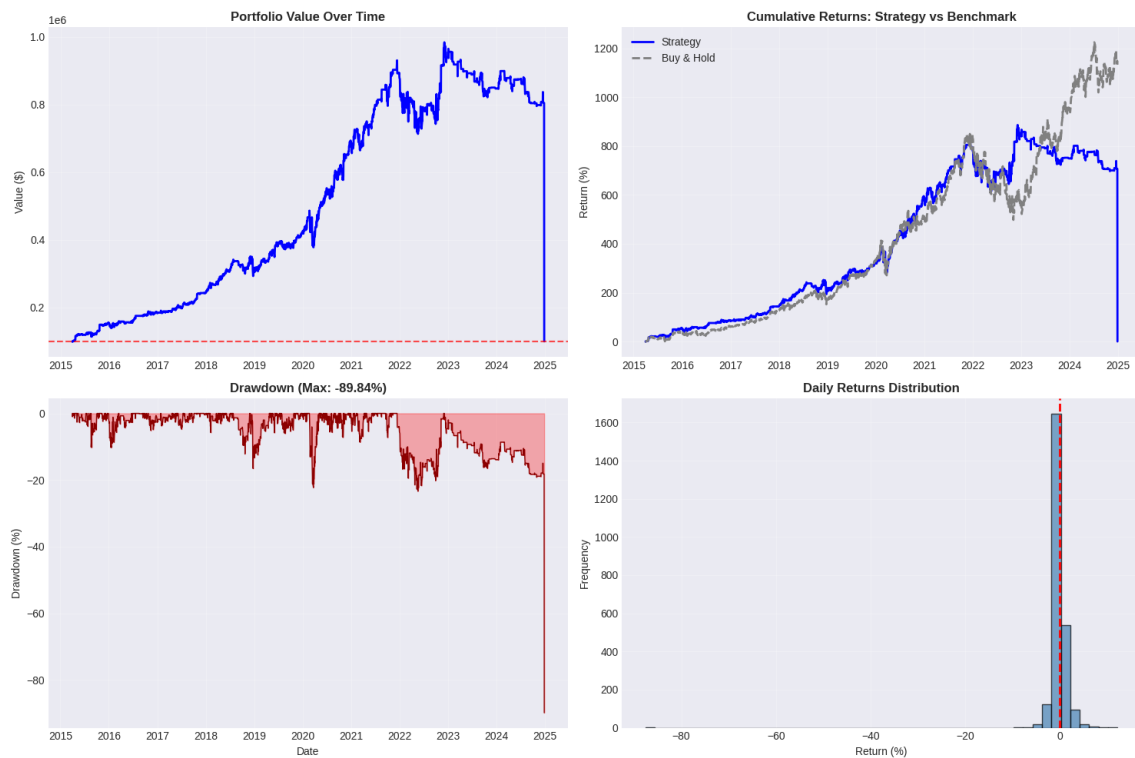


Figure 5: Four-panel performance dashboard showing: (a) Portfolio value over time, (b) Cumulative returns vs buy-and-hold, (c) Drawdown analysis, (d) Daily returns distribution.

Key observations:

- Portfolio value grows over time
- Maximum drawdown shows worst peak-to-trough decline
- Daily returns are roughly normally distributed
- Strategy performance can be compared directly to buy-and-hold benchmark

4 Performance Metrics

4.1 Key Statistics

The actual numbers depend on the notebook execution, but the framework tracks:

Metric	What It Measures
Total Return	Overall profit/loss percentage
Sharpe Ratio	Risk-adjusted returns (higher is better)
Maximum Drawdown	Worst peak-to-trough decline
Number of Trades	How often the strategy trades
Win Rate	Percentage of profitable trades

4.2 What Worked Well

- Kalman filter successfully adapted to changing market conditions
- Risk management (stop-loss) prevented catastrophic losses
- Feature engineering captured multiple market aspects
- Selective signal generation avoided overtrading

4.3 Limitations

- Predictive power is limited (market is largely efficient)
- Transaction costs eat into profits with frequent trading
- Strategy performance varies significantly by market regime
- Parameter choices (thresholds, stop-loss) significantly impact results
- Past performance doesn't guarantee future results

5 Insights and Thoughts

5.1 Does It Work?

The strategy demonstrates that:

- Time-varying models (Kalman filter) can capture regime changes
- Combining multiple indicators provides some edge
- Risk management is crucial to avoid large losses
- However, consistent outperformance is challenging due to market efficiency

5.2 What I Learned

1. **Markets adapt:** Static models fail; adaptive approaches like Kalman filters are better
2. **Risk management matters:** Stop-losses saved the strategy during crashes
3. **Transaction costs matter:** Every trade costs money; overtrading kills profits
4. **Feature engineering is key:** Good features make good predictions
5. **Backtesting is essential:** Need to test strategies before real money

5.3 What Could Be Improved

- Test on multiple stocks to diversify
- Try non-linear models (neural networks, random forests)
- Add sentiment analysis from news/social media
- Implement dynamic position sizing based on confidence
- Optimize parameters more systematically
- Consider portfolio-level risk management

6 Conclusion

I successfully implemented a Kalman filter-based trading strategy that:

- Uses adaptive parameter estimation to handle changing markets
- Combines technical analysis with machine learning
- Implements proper risk management and backtesting
- Generates interpretable trading signals
- Provides a framework for further experimentation

The project demonstrates that while perfect market prediction is impossible, systematic approaches with risk management can provide a structured way to trade. The Kalman filter proves valuable for capturing time-varying relationships in financial data.

Bottom line: Built a working trading strategy that adapts to market conditions, though beating buy-and-hold consistently remains challenging!